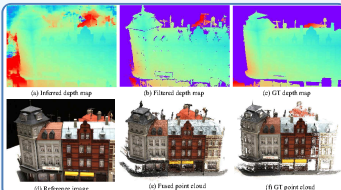


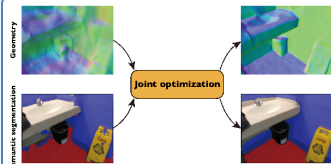
a Offline Reconstruction

MVSNet ECCV 2018
Cost-volume MVS



Images → Depth → Point Cloud

Manhattan-SDF CVPR 2022
Indoor neural SDF

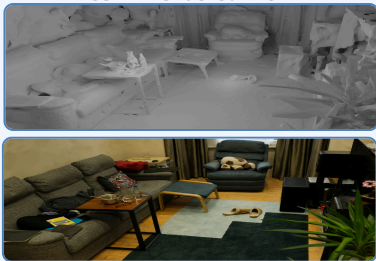


RGB + Semantics → Geometry

SuGaR CVPR 2024
Input Views



Mesh + Rendered View



Per-scene geometry optimization

b Feed-Forward Geometry

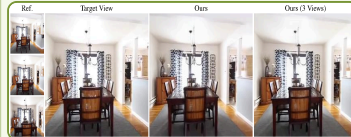
DUST3R CVPR 2024
Pairwise point maps + global alignment



Images → Aligned Point Maps

pixelSplat CVPR 2024

Feed-forward Gaussians

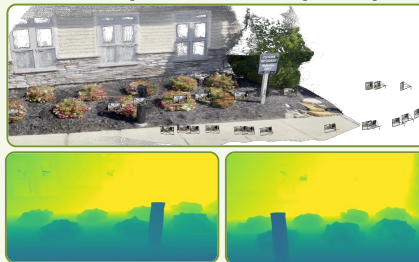


Views → Gaussians → Novel Views

VGGT CVPR 2025
Input Views



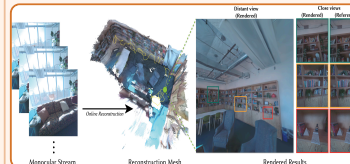
Point Map + Cameras / Depth Maps



Single-pass geometric inference

c Online Reconstruction

MODP SIGGRAPH 2025
Monocular online 3DGS mapping



RGB Stream → 3DGS Map → Rendered Views

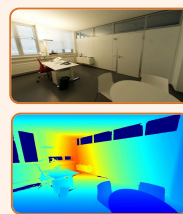
NICE-SLAM CVPR 2022
Hierarchical neural map



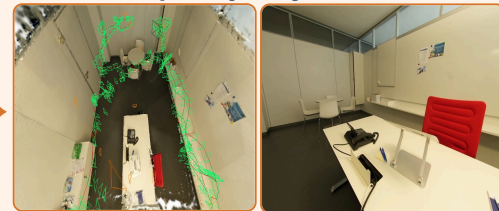
RGB-D Stream → Mesh + Camera Trajectory

SplaTAM CVPR 2024

RGB-D Stream



Gaussian Map + Trajectory / Rendered View



Streaming tracking and mapping